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Impact of Dataset Size and Class Diversity on ResNet Performance

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***Abstract*—In this study, we systematically investigate the impact of statically changing the dataset size and class diversity of the ResNet model to evaluate its performance. A Convolutional Neural Network is prominently used for image classification tasks. The experiments are conducted on the CIFAR-10 dataset, which is a standard benchmark for evaluating CNN models. The primary objective of this paper is to analyze the impact of varying the number of selected training samples and the range of classes included in the dataset to calculate models' performance using Accuracy and Macro F1 as standard parameters. We will take subsets of varying ranges and assign them with equal samples per class to check the effects of dataset size and class diversity on ResNet's ability to maintain high performance on unseen data.**

Index Terms— Convolutional Neural Networks (CNN), ResNet, CIFAR-10, Training Dataset Size, Class Diversity, Model Performance, Accuracy, Macro F1 Score.

I. Introduction

Before diving into a deep understanding of CNN, it is important first to understand its basics and origin. Deep learning, which forms the foundation for CNNs, is a type of machine learning [1], in which computers learn patterns and features from raw data by using multiple layers of artificial 'neurons', called a neural network. In contrast, older machines required humans to specify the basic functionalities and needed a step-by-step input explanation. That's when deep learning models were introduced, which can identify important features on their own, making them quite effective at handling data such as images, sounds, and text, where defining manual rules is difficult. This ability to automatically learn and extract features makes deep learning a powerful tool for solving complex real-world problems [2].

To get an understanding of how a basic CNN-based ResNet model works, refer to Fig. 1. The image is first fed through numerous convolutional filters, which are all set up to recognise particular features like edges, textures, or parts of objects. This in turn generates a grid-like mapping that specifies the unique features into a single component of the grid defined using numbers, then the pooling layers are applied to down-sample these maps, reduce spatial dimensions, control computation and forcing hierarchical abstraction. By stacking up several layers of convolutions and pooling, the network gradually pieces together more complicated patterns, so lines turn into shapes, shapes turn into objects, etc. At the end, the model "folds" all these feature maps into a one-dimensional vector. This vector forms a highly dense interconnected layers, which do the actual classification task of finally identifying the image. That's how the network finally picks the most likely label for the image.

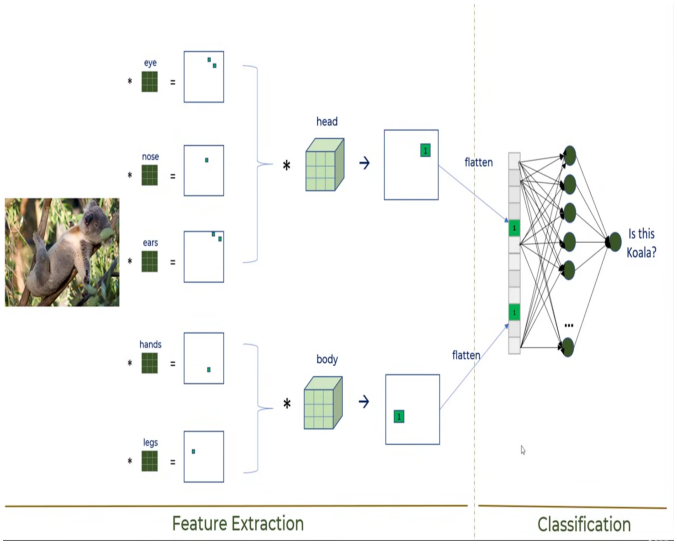

Fig. 1. Simplified illustration of feature extraction and classification in a CNN. Low-level features (eyes, nose, ears) are combined into high-level representations (head, body), which are flattened and passed to a classifier. Image adapted from the YouTube video : *"Simple explanation of convolutional neural network — Deep Learning Tutorial 23 (TensorFlow & Python)"* [3].

The development of Residual Networks (ResNet) by He et al. in 2015 was a major milestone in deep learning. Prior to ResNet, training deep neural networks had been difficult owing to the vanishing gradient issue [4]. Ideally, when the number of layers is increased, the model should be able to learn the identity mapping, meaning the network will not change the input and pass it forward untouched; however, CNN models struggled, which resulted in a drop in accuracy. ResNet resolved this challenge with its special skip connection feature, where inputs could skip one or more convolution layers, which helped in facilitating learning since the layers only have to learn the residual from the input and not the entire thing.

Among the various ResNet variations, ResNet18 is a notable standout in research. It strikes a sweet spot: powerful enough to pick up complex image features, yet not so heavy that the training drags or chews up memory. That is why it

provides the best value when researchers want to test how changing the size or diversity of the training data affects certain metrics like Accuracy or Macro F1 Score. Running these tests with ResNet18 gives us a much sharper look at limited data or lots of classes regarding how deep models handle these conditions. It is actually a real window into model robustness and generalisation.

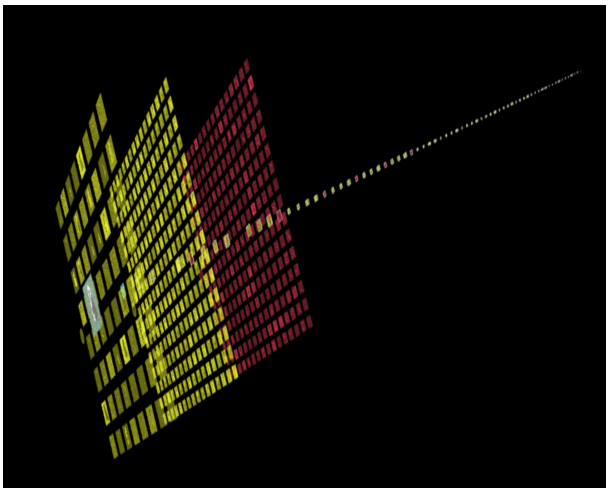

Fig. 2. Simplified illustration of a Residual Network (ResNet) with a lot of convolution layers showing a skip connection [5].

II. Related Work

Recent studies have investigated how the size and diversity of Training data impacts the performance of deep neural networks. Building on this basis, we shall examine the impending impact of dataset size and class variation applied to the ResNet models using accuracy and macro F1 scores as major evaluation measures.

In order to show a direct correlation between training sample size and performance metrics like accuracy, Shahinfar et al. [6] examined the sample size needed for deep network classifiers with regard to long-term wildlife monitoring.

Zhao et al. [7] gave a comprehensive survey on the combined problem of small sample sizes and class imbalance. Proposing a single analytical framework and measures for evaluating model robustness under such conditions.

Davidian et al. [8] discussed the relationship between the dataset size, class imbalance, and CNN performance in the medical field. They investigated the influence of changing the number of training samples on diagnosis accuracy, which was influenced by class distribution, by using a chest X-ray dataset for COVID-19 detection.

Chen et al. [9] presented a comprehensive survey on imbalance-learning, underlining the challenges created by biased class distributions in machine learning and deep learning, also proposing strategies and best practices to improve the model performance in different applications.

Rajput et al. [10] studied how the number of training examples can change the outcome of a machine learning model. They found that looking at both how much the model's predictions vary and how accurate they are gives a better sense of whether the dataset is big enough to trust the results.

The authors, Hlynsson et al. [11] present a framework to test the effect of the size of the training dataset on deep learning models. They compared CNNs on the MNIST and Omniglot datasets with a model called HiGSFA, examining how different amounts of training data affect performance.

Rezvani and Wang [12] helped us by providing a thorough analysis of class imbalance learning strategies by dividing them into three categories: hybrid approaches, algorithmic structures, and data pre-processing, while also assessing the methods' performance in both regression and classification-related tasks.

Shwartz-Ziv et al. [13] demonstrated that models can function well even in cases where classes are unbalanced. This was achieved by carefully modifying parameterised settings of batch size and data augmentation.

A thorough analysis of the ResNet architectures' capacity to generalise under various training circumstances was carried out by Du et al. [14]. Their research methodically examined how variables like sample diversity, class distribution, and dataset composition affect the stability and predictive capabilities of ResNet models.

In this study, we investigate how different training dataset sizes and class compositions impact ResNet model performance. Using the CIFAR-10 dataset as a benchmark, we can evaluate the model's learning behaviour and adaptability. The aim is to better understand how convolutional networks perform when faced with limited data and uneven class distributions, which are mainly encountered in real-world applications.

III. Consider Model

Let's strive to understand the residual network's tremendous importance in deep learning. In a regular deep neural network, each input, let's say x, generates an output H(x) that progressively learns different aspects of the input item. However, adding more layers to this typical deep neural network led to major problems. Although it would seem that this would improve model accuracy and learning capacity, in reality, deeper networks frequently performed worse. This occurs as a result of the deeper layers' inability to "pass the useful information forward" or learn identity mappings.

Instead, the data was distorted, which increased the complexity of the layers when they were combined. By integrating skip connections, ResNet transformed deep networks and made it possible for far deeper architectures to train efficiently. It taught each layer to simply learn what needed to change given the input x, rather than requiring each layer to learn the entire transformation H(x).

$$H(x) = F(x) + x \tag{1}$$

In the above equation, F(x) stands for the modification or adjustment that the network must make to x.

As a result, the model no longer needs to continuously relearn identity mappings since the layer just learns what is missing or needs to be fixed, rather than learning a whole transformation from the beginning [15]. One of the main vanishing gradient issues was also resolved by ResNet. When learning deep neural networks, we backpropagate the gradient flow from the output to the input. It was found that as gradients were multiplied through the derivative of each layer, they became smaller and smaller until they vanished, halting the learning process for the first few layers. In order to prevent gradient flow from being diminished by all the intermediate layers, skip connections are used to construct shortcuts in backpropagation.

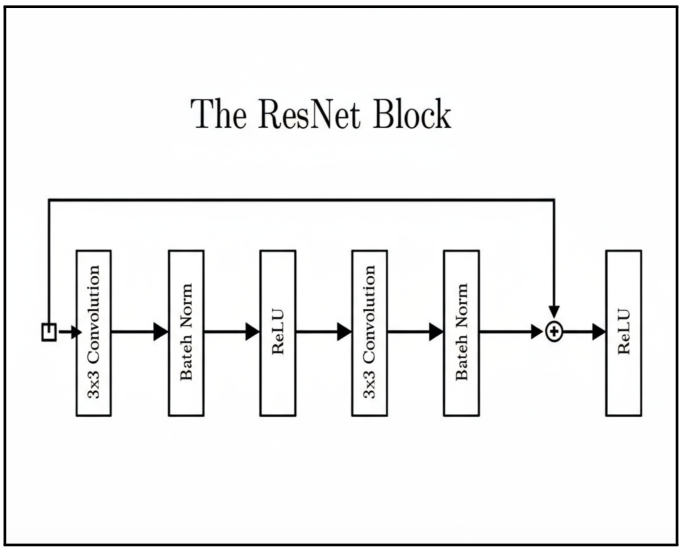

Fig. 3. Block diagram of a Residual Block illustrating the skip connection. The image was AI-generated, inspired by the ResNet block structure presented in *"3DPalsyNet: A Facial Palsy Grading and Motion Recognition Framework Using Fully 3D Convolutional Neural Networks"* [16].

Residual Block Architecture

The fundamental component of the ResNet model is the *residual block*, illustrated in Fig. 3. Instead of learning an entire transformation, each block focuses on learning the *residual mapping*, which refers to adjustments in the network that convert the input into the desired output. This design greatly improves gradient flow and stabilises the training of very deep networks.

1) **Input:** The input feature map enters the block and is processed sequentially through the following layers.
2) **3×3 Convolution:** Applies a small 3×3 kernel to extract local patterns such as edges and textures from the input image.
3) **Batch Normalization:** Normalizes the activations to maintain a stable distribution, reducing the risk of vanishing or exploding gradients and improving training efficiency.
4) **ReLU Activation:** Helps the network go beyond simple rules and learn complex hierarchical feature relationships beyond simple linear mappings.
5) **Second 3×3 Convolution:** Further processes the features, enabling the model to capture deeper and more abstract representations.
6) **Second Batch Normalization:** Stabilizes the output of the second convolution before combining it with the skip connection to ensure numerical consistency.
7) **Skip Connection (+):** Adds the original input directly to the block's output, preserving essential information and preventing gradient degradation during backpropagation.
8) **Final ReLU Activation:** Applies ReLU after the addition to propagate only the most informative combined features to the next block.

IV. Experimental Setup

For all those curious readers, it's important to understand the methodology and all the backend work done in order to conduct this research paper.

We set up our experiments using the CIFAR-10 dataset to see how well the model performs when we limit the data in different ways. Everything ran in Google Colab, so we took advantage of its GPUs to train and test ResNet18 quickly.

A. Data Preparation

Using the CIFAR-10 dataset [17], which contains 60,000 colour images of (32×32 resolution), which are evenly distributed across ten classes, namely Aeroplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, and Truck sequentially labelled from 0-9 respectively. Without worrying about class imbalance or changing the total sample count by keeping things consistent, we started with 1,000 samples for each class by simply adding more classes one at a time and preventing uncontrolled piling of data.

B. Model Architecture

We employed ResNet-18 as the base model due to its proven performance and efficiency in image classification tasks. The model parameters and architecture were kept constant across all experiments to ensure fair comparison. Furthermore, ResNet models are very efficient in transfer learning. Salient features can be recovered and reused across tasks via pre-trained weights, which can significantly help in reducing the computational cost of training the model from scratch while maintaining good accuracy in generalisation.

C. Training Procedure

The ResNet18 model was trained in PyTorch for 5 epochs using Stochastic Gradient Descent (SGD) on the CIFAR-10 dataset. We kept our learning rate set to 0.01, the momentum to 0.9, and the batch size to 64. All images were resized to 224×224 pixels in order to match the input size expected by the ResNet model.

A batch size of 64 was chosen to maintain a good balance between training speed and stable gradient updates, where

generally it is observed that smaller batches can make training noisy and unstable; conversely, very large batches can lead to overfitting. The learning rate of 0.01 allowed the model to learn efficiently without causing unstable updates. Training for 5 epochs provided enough learning for comparison while maintaining a reasonable computational time.

D. Performance Evaluation

We measured model performance with two main metrics: Accuracy and Macro F1 Score. While accuracy aids in measuring overall correctness, it does not show how uniformly the model performs across individual classes, particularly when there is a subsequent increase in class diversity. This is where MacroF1 comes in; it demonstrates whether the model's capacity to treat all classes fairly declines as the number of classes increases.

V. Result and Discussion

Table 1 displays the performance matrix of the ResNet18 model across seven experimental configurations, including the full CIFAR-10 dataset as a baseline.

Accuracy measures how often the ResNet18 model predicts the correct class across all selected classes and samples in the experiment. However, a high Accuracy does **not** guarantee good model performance in the presence of class diversity.

Although the dataset is balanced equally across individual classes, MacroF1 helps measure the per-class performance equally regardless of the model's average. This helps identify whether ResNet generalises uniformly across all classes or exhibits biases toward specific visual patterns.

In conclusion, the Macro F1 score provides more detailed information about the ResNet18 model's class-wise consistency, whereas Accuracy measures the model's overall accuracy. When combined, these measures offer a thorough assessment of the model's overall effectiveness and its capacity for consistent generalisation across various classes.

TABLE I
Calculating Performance on Accuracy and Macro F1 scores

Class Diversity	Dataset Size	Accuracy (%)	Macro F1(%)
[0, 1, 2, 3, 4]	5,000	60.44	60.14
[0, 1, 2, 3, 4, 5]	6,000	53.43	52.46
[0, 1, 2, 3, 4, 5, 6]	7,000	50.89	49.71
[0, 1, 2, 3, 4, 5, 6, 7]	8,000	49.83	48.91
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]	50,000	66.47	66.44

One key question that arises from Table I is why, despite training on all ten categories of CIFAR-10 with a total of 50,000 training samples, the model fails to achieve an accuracy of 90–100% on the full dataset, while it can reach only around 66%. This outcome is primarily due to the model being trained for a relatively small number of epochs (*epoch* = 5). An epoch refers to a single complete pass of the entire dataset through the network. With a low number of epochs, the model is not fully exposed to the whole dataset, which restricts its ability to fully learn the underlying patterns and representations. Conversely, increasing the number of epochs to 50 or 100 would likely lead to higher accuracy.

However, achieving higher accuracy would come with a cost of increased computational time and memory allocation. Therefore, for research purposes, it was a deliberate decision to keep a low epoch to overall understand the accuracy and efficiency of our model.

Observation 1: Small gap between Accuracy and Macro F1 across all subsets

At 5,000 samples, we can observe that the gap is 0.30. Similarly, at 8,000 samples, the gap between accuracy and Macro F1 scores is found to be around 0.92%.

From Observation 1, it is evident that the model performs consistently across all classes. Because we kept the dataset balanced with 1000 per class sample, there are no signs of bias shown by the model towards any class, henceforth ensuring that each class contributes equally to the overall Accuracy. This demonstrates fair and uniform performance without bias toward any specific class.

These observations can be clearly represented and demonstrated via the line graph given below :

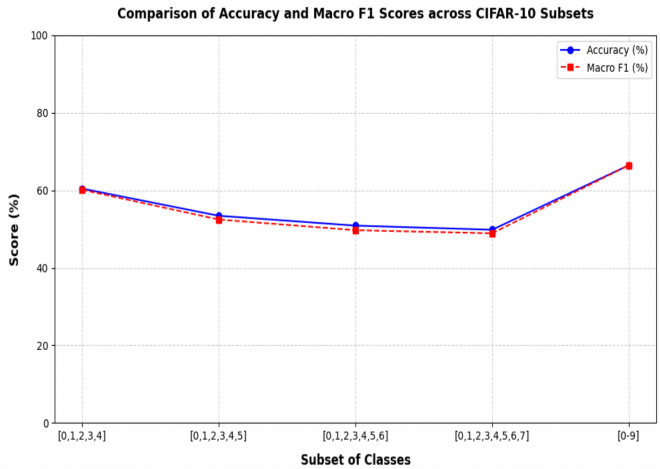

Fig. 4. Comparison of Accuracy and Macro F1 across different CIFAR-10 subsets.

As shown in Figure 4, the two lines nearly overlap, indicating the strong efficiency and stability of the ResNet18 model. The close match shows that the model performs consistently across all classes and does not show any significant bias. These observations agree with the findings of [18], who also reported similar performance uniformity across classes using ResNet18.

Observation 2: Higher Accuracy and Macro F1 with small class diversity and small dataset

Despite the small sample size, each class received 1000 samples, which helped every class to learn promptly and correctly. Furthermore, fewer classes made it easier for the model to learn without much inter-class confusion, also reducing feature overlap between categories. This reduced complexity for the ResNet model and generated higher performance parameters for accuracy and F1.

Observation 3: Accuracy and Macro F1 decrease as dataset size and class diversity increase

It was shown that both F1 and Accuracy scores decreased as dataset size and class diversity increased, with Accuracy falling from 60.44% to 49.83% and Macro F1 from 60.14% to 48.91%. This is mainly due to the increased complexity of the model. When classes have similar features, such as cats and dogs, it becomes harder to identify distinct visual traits. As more classes are added, the learning challenge grows faster than the advantages of a larger dataset.

Fig. 5. Comparison of Accuracy and MacroF1 across different `CLASS_DIVERSITY` subsets and the complete dataset.

Taking the full CIFAR-10 dataset, comprising classes 0 through 9, for comparison provides a point of reference for assessing the model's generalising capability with complete class diversity and maximum available data. The baseline shows the highest level of complexity the model must deal with, since all ten categories are covered, with the full 50,000 training samples included. In this way, by comparing subsets with less class diversity and smaller dataset sizes to this complete dataset, changes in model performance can be appreciated as task difficulty increases.

The precision and overall accuracy of the model gradually decrease as the number of classes and dataset size increase, as shown in Figure 5. This trend is especially clear in the early phases, when accuracy drops sharply from about 60% to 48%, as seen in Figure 4. The main cause of this initial sharp decline is that adding more classes leads to considerable confusion between them. The wider variety of visual characteristics forces the ResNet18 model to broaden its decision limits, leading to misclassifications and a marked drop in performance.

These findings align with those of Dablain et al. (2023), who observed similar sensitivity of CNN-based models to increasing class diversity and dataset imbalance, emphasising that model generalisation tends to degrade with rising variables and overlapping feature distributions [19].

VI. Conclusion

With this, we are close to fully wrapping up our study and can conclude the following findings:

- Nevertheless, the entire dataset (classes 0–9, 50,000 samples) shows a discernible improvement in Macro F1 and Accuracy. The greater feature diversity that improves generalisation and the improved data availability per class are responsible for this recovery. The ResNet18 model gains a richer and more representative feature space when trained on the entire dataset, which improves its ability to differentiate between visually similar categories. The final inclusion of all 10 classes allows the model to learn more robust, discriminative representations across the entire data distribution. The findings of Hestness et al. (2017), who showed that model generalisation and performance continuously improve with increased dataset scale and variety [20], are consistent with this fact.
- Training the model for only 5 epochs restricts its ability to fully converge. The network does not have enough iterations to optimise weights and effectively minimise loss. Afaq and Rao [21] state that a lower number of epochs often leads to incomplete learning. This prevents the model from capturing complex data patterns and reaching its highest accuracy. While 5 epochs offer a quick and steady baseline for comparing subsets, the shorter training time limits the model's overall performance.
- Additionally, we utilised a constant number of samples per class in the formation of our training subsets to guarantee uniform representation across all categories. The method is selected to make the class variety an independent variable and not be influenced by the imbalance of the dataset, so as to provide a controlled test of the generalisation capability of ResNet18. By keeping an equal count of samples for each class, we avoided confounding influences due to different class sizes and ensured that any differences in performance were due exclusively to the number of distinct classes involved. This technique helped us determine that class diversity and subset size are critical factors in determining model performance.

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